

Yuxuan Liu

AGENT INTERN, EVALUATION & ENGINEERING · AI DATA & SAFETY

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Summary

Full-stack engineer for multimodal Agent systems oriented toward AI-native applications. Independently designed and shipped **Drawsee**, a complete AI Agent platform for circuit-design education — integrating an original closed-loop workflow, agent-trajectory collection, and canary-based self-optimization infrastructure. Currently preparing an academic submission, under advisor guidance, on a post-training method incubated from this scenario.

Education

Beijing University of Posts and Telecommunications (BUPT)

Beijing, China

B.ENG. IN TELECOMMUNICATIONS ENGINEERING (IN PROGRESS)

Class of 2027

Core Project Experience

Drawsee (昭析) — A Multimodal AI Dialogue & Circuit-Analysis Platform for Electronics

drawsee.cn

Education

FOUNDER & BACKEND LEAD (PERSONAL PROJECT)

Mar. 2025 – Present

- Sole backend owner from the first commit to today: independently designed the system architecture and built/iterated on every core backend module — Git history shows over **330 commits in the past 18 months, more than 90% of the repository's total**. Tech stack: Spring Boot 3.4 + LangChain4j + Qdrant + MySQL + Redis + RabbitMQ + MinIO.
- **Circuit-Design Agent Closed-Loop Workflow** — designed from scratch a closed-loop agent architecture, “Plan → Generate → Structured Verification → Repair → Archive,” that turns vague circuit requirements into structured specs and interface contracts, uses a *deterministic simulator* (rather than model self-evaluation) for functional-level verification, and automatically enters a rule-based repair-and-reverify loop on failure; abstracted the loop's termination policy into stateless pure functions so core decisions — when to stop, when to escalate to a human, when to declare failure — can be tested and evolved independently of the sandbox and the LLM.
- **Drawsee Self-Improvement System** — led the design and implementation of a fully automated self-optimization loop covering “issue discovery → candidate generation → canary validation → auto-promotion/rollback”: user feedback is auto-clustered into optimization candidates, gated by review, then canary-released; effectiveness is judged with *statistical significance testing* (a two-proportion sequential test) instead of manual judgment. Fixed three statistical/attribution defects in the prior implementation that had been producing incorrect conclusions, giving the product continuous self-iteration without manual case-by-case review.
- **In-House Agent Evaluation Harness (TestLab)** — designed and built the team's general-purpose Agent evaluation infrastructure from zero: custom orchestration for eval tasks, environment isolation and credential management, and a *deterministic-replay* evaluation engine that lets any Agent decision be replayed step by step, fingerprint-compared, and traced to the exact point of deviation — turning “is this eval result trustworthy” into an engineering property that tests can guard long-term. Also built the structured recording/export pipeline for Agent execution traces, turning live interactions into data assets directly usable for model training.
- **Backend Foundation Architecture** — independently owned technology selection and platform build-out for the entire backend: unified orchestration and dynamic context-budget management across multiple models (Qwen / DeepSeek), a RAG retrieval pipeline (Qdrant vector store + hybrid retrieval), RabbitMQ async tasks with multiple worker types, Redis Stream + SSE streaming, MinIO object storage, Flyway versioned migrations, Sa-Token auth, and multi-environment CI/CD — laying the foundation the team later built on for the self-improvement system, evaluation harness, and more.

Research & Competitions

ONGOING RESEARCH

FORGE: Verifier as Teacher

IN PROGRESS — TARGETING IJCAI-27 / AAAI-28 2025 – Present

- Working with my advisor, incubated a post-training method from Drawsee’s circuit-design agent scenario: in engineering-verification environments that support counterfactual queries (e.g. EDA design, GPU kernel optimization), a deterministic verifier acts as a “counterfactual teacher,” constructing *Verified Counterfactual Pairs* (VCP) for model-proposed design edits and distilling verifier-guided search and effect knowledge into a small model’s weights — so the student model achieves higher task success and better effect-prediction calibration on unseen tasks/circuit families **without** needing extensive, high-latency tool calls at deployment. Currently in mid-stage experimental validation.

TraceParse: A Candidate-Graph Repair Framework for Schematic Parsing

IN PROGRESS — TARGETING CVPR 2027 2025 – Present

- Focused on schematic-to-netlist parsing: identified that the dominant bottleneck is not recognition accuracy but the **structural unrepresentability** of the candidate connectivity graph — the correct netlist simply does not exist anywhere in the candidate space, so improving recognition accuracy alone cannot fix it. Proposed a “repair-then-parse” framework: a library of deterministic repair operators first makes the candidate graph representable, then an exact parsing algorithm solves it, with machine-checkable structural certificates flagging genuine conflicts. The repair module is decoupled from the underlying image perceiver and transfers across vision-language models of different scales. The paper’s core results are complete, targeting a CVPR submission in the coming months.

HONORS & AWARDS

2025	2nd Prize , China International College Students’ Innovation Competition, a.k.a. “Guochuang Cup” (Beijing Region) — Drawsee	Beijing
2024	1st Prize , China Collegiate Computer Design Competition (Beijing Region) — Drawsee	Beijing
2024	2nd Prize , National University-Industry ICT Education Integration Competition — Drawsee	
	1st Prize , 10th National College Physics Experiment Competition (Individual)	

Skills

AI / Agent Engineering	Agent-trajectory evaluation & deterministic replay , LangChain4j agent tool orchestration & tool calling, structured-output design (Spec/Plan/Contract system), RAG (retrieval-augmented generation)
Post-Training Algorithms	SFT (supervised fine-tuning), GRPO (Group Relative Policy Optimization), OPD (On-Policy Distillation) — principles and engineering implementation of mainstream post-training paradigms
Languages	Java (primary, 2+ years of engineering practice), Python, Golang (hands-on experience with mainstream frameworks on side projects)
Backend & Middleware	Spring Boot, MyBatis, MySQL (Druid + Flyway), Redis (Redisson), RabbitMQ, MinIO, Qdrant, Sa-Token, FastAPI, Gin/GoFiber
Engineering Practices	Git, GitHub Actions CI/CD, JUnit + Testcontainers integration testing, technical design docs